

Meghana Baladari

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Summary

Master's student in Data Science with experience building end-to-end data and machine learning workflows across structured and unstructured data. Skilled in Python, SQL, data modeling, feature engineering, pipeline development, model evaluation, NLP, and AI systems. Seeking data science, analytics, and data engineering opportunities involving ETL, scalable data systems, and data-driven decision-making.

Education

Arizona State University - M.S. in Data Science, Analytics & Engineering

May 2027

University College of Engineering, Osmania University - B.E. in Artificial Intelligence & Machine Learning

2021 - 2025

Professional Experience

Vosyn - MLOps Engineer Intern

May 2026 - Aug 2026

- Evaluated **3 pretrained Cantonese TTS and voice-cloning models** (Qwen3-TTS 0.6B, WSYue-TTS, CosyVoice2-Yue), comparing speech generation quality and practical usability across GPU environments.
- Set up model environments and resolved **4+ dependency and runtime conflicts** involving outdated ONNX Runtime, PyTorch, and TensorRT version pins to configure pretrained checkpoints for reproducible speech generation experiments.
- Ran inference and reference-voice cloning experiments, achieving **100% generation success** across test cases with near real-time latency (**RTF ~0.94**), and validated outputs for stability, intelligibility, pronunciation, and speaker similarity.
- Investigated Cantonese-specific evaluation methods, including Jyutping Error Rate and Tone Error Rate, as part of a **5-metric evaluation framework** built to assess pronunciation and tone accuracy across models.

Solix Technologies - AI Intern

Oct 2024 - Dec 2024

- Built an end-to-end NLP content-generation pipeline using **Python, Hugging Face Transformers, and OpenAI APIs**, integrating preprocessing, model inference, and prompt-based generation.
- Engineered a **3-stage NLP preprocessing pipeline** with NLTK and spaCy for tokenization, lemmatization, and stopword removal across domain-specific text.
- Fine-tuned transformer models on domain-specific datasets, improving generated-content consistency by **18%** over the initial baseline.
- Documented system architecture, API workflows, and prompt engineering strategies for scalability.

Coincent - Intern

Apr 2024 - Jun 2024

- Developed an end-to-end computer vision pipeline using **TensorFlow/Keras and the CIFAR-10 dataset**.
- Built preprocessing, normalization, augmentation, batching, and training workflows for large-scale image data.
- Performed hyperparameter tuning and model evaluation across multiple experimental configurations.
- Achieved **85% Top-1 accuracy and 95% Top-5 accuracy**, outperforming the baseline CNN by **12%**.

Projects

Imbalanced Binary Classification with XGBoost - Python, Scikit-learn, XGBoost, Pandas

Fall 2025

- Built an end-to-end classification pipeline on **10,000 samples and 21 features**, addressing an approximately **3:1 class imbalance** through stratified splitting and class weighting.
- Trained and tuned an XGBoost model using **300 estimators**, imbalance-aware weighting, and Balanced Error Rate metrics to improve evaluation of the minority class.
- Achieved **63.18% balanced accuracy** and approximately **73% overall accuracy** on unseen test data, demonstrating a more reliable evaluation than accuracy alone for imbalanced data.

Air Quality Sensor Calibration using Machine Learning - Python, Gradient Boosting, Random Forest

Fall 2025

- Built calibration models for **5 air pollutants** using low-cost sensor and meteorological data.
- Engineered **1-, 6-, 12-, and 24-hour lag features**, rolling statistics, and temperature/humidity interactions to capture temporal and nonlinear sensor behavior.
- Compared **3 model families** - Ridge Regression, Random Forest, and Gradient Boosting - using time-aware validation, with Gradient Boosting delivering the strongest performance across pollutants.

Online Placement & Networking Portal - React, Node.js, Express, REST APIs

Fall 2024 - Spring 2025

- Built a full-stack student networking platform with profiles, posts, project listings, and collaboration requests, supported by **21 developed and tested backend APIs**.
- Implemented JWT-based authentication with **Supabase/PostgreSQL** to manage secure user sessions and protected application workflows.
- Integrated **GPT-4o through the OpenAI API** as an AI writing assistant, combining full-stack development with generative AI to help users create professional posts.

Spelling Correction Tool - Python, NLTK, N-Gram Modeling, Jupyter Notebook

Spring 2024

- Built a **3-stage NLP spelling-correction pipeline** using edit-distance candidate generation and contextual bigram ranking.
- Implemented unigram + bigram language models with NLTK to improve correction decisions using local word context.
- Evaluated the model on **100 test sentences**, achieving **80%+ reported correction accuracy** on common misspellings.

Technical Skills

Programming & Querying: Python, SQL, C++, Java

Data & Analytics: Exploratory Data Analysis, Statistical Analysis, Data Modeling, Data Visualization, Feature Engineering, Time-Series Analysis

Machine Learning & AI: Machine Learning, NLP, Deep Learning, Computer Vision, Transformers, Model Evaluation

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Hugging Face Transformers, NLTK, spaCy

Data & Development Tools: Git, GitHub, Jupyter Notebook, Google Colab, Anaconda, PostgreSQL

Certifications

IBM - Machine Learning with Python

IBM - Databases & SQL for Data Science with Python

Meta - Programming with JavaScript

OneRoadmap - AI Engineer